2026-10-07: a mods round for capability item 1, on 2.1.293, before any function hook is planned on

This extends item 1 of the capability lane in the report-v5 build queue (handoff 0cw, "Host measurement"). It adds measurement and builds nothing. It is handoff material, not roadmap text: no direction changes, and the roadmap and the queue keep their words. Anything below that would change direction waits for the founder's yes.

Why. Claude Code 2.1.293 documents mods: plugins whose hooks are typed functions that run inside the session (`turn.step`, `agent.spawn`, `tool.call`, `session.measure` and many more events), where a hook used to be a shell command that reads stdin. A built-in plugin-authoring plugin first shows in the init lines of 2.1.286 and not in 2.1.284 or 2.1.260 (host-measurements `results.md`). Item 1's measurements covered command hooks: every item on 2.1.260 and 2.1.284, and a decisive subset on 2.1.286. The queue rests on several things a hook "cannot do". The 2.1.293 declarations say otherwise for function hooks. A declaration is a claim about a build and not a measurement of it, and the declarations' own header says the surface may change between releases without notice. So this round measures first, as item 1 did.

Premises that may move. Every right-hand cell is declared, not measured.

Premise the queue holds	Where	What the 2.1.293 declarations say	Q
No hook can set the main session's model or effort	review §3	`turn.step` lets a hook send another `model` or `effort` on each request, for the main thread and for subagents	2, 3
Writing a floor model into an Agent call can lower the model, so the floor stays advisory	0cw lane 8; wave 1	`agent.spawn` gives the hook the parent's model (pinned) and returns the resolved `model` and `agentId`	5
`--max-budget-usd` is soft: 39 of 39 capped runs exited 0 over the cap	0cw wave 1	a `turn.step` answer without `next` sends no request, and `session.measure` pushes cost	6
Under `--restricted` only plugin hooks fire, and `--bare` fires none	results M2	a mod is a plugin; whether its function hooks load under either flag is not declared	1
91% of subagent traffic is Workflow agents	review §4.2	`agent.spawn` can only refuse a Workflow agent; whether `turn.step` fires for its requests is not declared	4
Felix's hooks fail open	0cw, the declined input-rewriting floor	a failing hook is skipped unless its `.catch` answers in its place	7

The questions. Each gets MEASURED-YES, MEASURED-NO or NOT-MEASURABLE, and the row says what would make a NOT-MEASURABLE measurable.

1. **Does a mod load?** A probe mod through `--plugin-dir` under `claude -p`, on 2.1.286 and 2.1.293: plain, `--restricted`, `--bare`, and `--setting-sources ""`. Also `claude plugin validate` and `claude plugin test` on the probe. *Decides:* whether a mod can sit on a lease's enforcing path at all.
2. **`turn.step` on the main thread.** Rewrite `model` to another full id, and rewrite `effort`. Read the request the stand-in received (it already logs `model` and `effort`), then what `resolvedModel` and `modelUsage` report. *Decides:* the first premise.

3. **`turn.step` in a subagent.** Is `agentId` present; does a rewrite reach the subagent's requests and not the parent's; what a `fork`, which inherits the parent's model, does.
4. **A Workflow agent.** This is the "Workflow-spawned agents" measurement already queued under item 1 ("Before step 8, measure"), so one scenario answers both. Does `agent.spawn` fire (it can only refuse), does `turn.step` fire with an `agentId`, does a rewrite take. If the Workflow tool is not offered under `-p`, the row says NOT-MEASURABLE and that it needs an attended Desktop run. *Decides:* whether mods matter for routing at all, since this is 91% of the traffic.
5. **`agent.spawn`.** The Agent call that names no model (the case measured to lower it). The hook raises, leaves and lowers. Compare three readings: the hook's returned `model`, the stand-in's `model` and `PostToolUse(Agent).resolvedModel`.
6. **A request-boundary stop.** Script the stand-in's usage, read cost from `session.measure`, answer `turn.step` without `next` past a threshold. Run the 39-of-39 shape (subagents lagging the main turn, `delay_ms`, `early_start`) and count the requests that still reach the stand-in after the stop. A stream already started is not stopped; count those apart.
7. **Failure.** A hook that throws, one that outlasts its budget, and one whose `.catch` denies. Do requests still go out. *Decides:* whether a mod fails open, as Felix's hooks do, or closed.
8. **Desktop, attended and optional.** A probe mod in the Desktop Code tab on 2.1.293: does it load, and what the one hot-reload question says. About ten minutes of the founder's, and the restart-moments probe visit can carry it. The probe draws nothing on screen.

Not measurable on the stand-in, and said so rather than guessed:

- **The cache cost of switching a model.** The stand-in has no cache. The review's number (a parent switch writes a median 321,582 cache tokens against about 1,100, n=12) stands. A mod that switches the main model is not run on real tokens before the replay decides.
- **Rate-limit windows.** `session.measure` leaves them empty off a subscription. A passive logger in the founder's own session would read them at no extra spend, if he chooses to enable one.

Method.

- **Rig.** Copy `settle2/rig` into `audits/2026-10-07-mods-round/` in the memory root, add `desktop-2.1.293` to the binary table (found under `~/Library/Application Support/Claude/claude-code/2.1.293/`), and record `--version` and sha256 at run time. The app re-lays its directories: the 2.1.284 path moved mid-measurement last time. Models are full ids, read from the host.
- **Isolation unchanged.** `env -i`, a scratch HOME, the stand-in at `ANTHROPIC_BASE_URL`, a dummy key, and `DISABLE_TELEMETRY=1` with `CLAUDE_CODE_DISABLE_NONESSENTIAL_TRAFFIC=1`. No credential is read and no token is spent.
- **Probe mods.** Deterministic, no model, no network, no UI calls. They live in scratch and load through `--plugin-dir`. They are not installed, and nothing is written under `~/.claude/plugins/cache/`.
- **Predictions first.** Before any run, a blind author writes the expected outcome of each question from a frozen copy of the declarations file kept with its sha256 (the plugin-authoring skill writes its copy under `/private/tmp` as it loads, and a restart names a new folder). Runs are judged against those predictions,

and a declared-but-wrong behaviour is a finding. Then one adversarial challenge round, as rounds 1 and 2 had.

- **Versions and load.** Each verdict is stated for 2.1.286 and 2.1.293. Timing verdicts, Q6 above all, use child CPU and not wall time, and carry the load average, which has run at 100 to 370 on this Mac.
- **Record.** Rows in `results.tsv` marked `[mods-round 2026-10-07]`, a section in `results.md`, and a `disposition.md` after the challenge round.
- **Stage models.** Predictions and the challenge round on Opus 5.5. Rig and probe plumbing, the runs, and the fact-check of the results on Sonnet 5.5 at high. A stage that goes red twice moves to Opus.

Not in this round.

- No engine change and no felix-home PR. Nothing touches `verify.sh`, `receipt.sh`, merge or the gate.
- No real tokens and no credential.
- No routing, floor or stop is built. The roadmap and the queue are unchanged.
- Nothing measured here becomes an input to a receipt or a merge blocker (trust invariance), and no probe puts a model on an enforcing path (invariant 2).

Size and when. About 0.5 to 1 lane session and no tokens. Spec and measurement work is allowed now: it is not an engine branch, so the "more than 10 engine PRs" rule does not stop it. It is not a suite or a gate, so it asks the carrier for no slot. It runs before item 4 (the dry-run lease) and item 6 (the outcome record), because the answers to 1, 3 and 6 change what those two choose.

What it may put in front of the founder. None of these is asked now.

- **If Q6 holds:** the attended tick could stop before the next request goes out past the cap, where today it is checked afterwards. The cost is that the lease would load one plugin, and the 2026-10-01 amendment says no plugin. His call, on the numbers.
- **If Q5 shows a typed spawn never lowers a model:** the declined input-rewriting floor can be reconsidered with evidence. It would still be an architecture change, an enforcing output where Felix's hooks fail open, so it stays unbuilt until he decides.
- **If Q3 and Q4 show `turn.step` reaches Workflow agents:** the review's 12.8% ceiling for moving Opus subagent work to Sonnet becomes reachable at the host and not only by editing scripts. The replay's predeclared rule still decides whether routing is built at all.
- **Anything a mod draws on screen** is the stealth question again. The model reminder stays chat-only.